NUMERICAL METHODS

Numerical Interpolation

Advanced Applications

Objectives

•To understand the concept of interpolation and its role in estimating values between known data points in real-world contexts.

•To enhance problem-solving and critical thinking skills by formulating real-world word problems into computational interpolation tasks.

Language

Up to you (Python / C++)

Task

A graphics designer is working on generating a symmetrical face outline for an animation. Due to data corruption, only half of the outline (left side of the face) is available. The coordinates of the half-outline are given at equally spaced intervals along the vertical axis (y-axis):

y(position along face height)	0(chin)	2	4	6	8	10
x(distance from midline, left side)	0.0	1.1	1.8	2.4	2.0	1.3

- Using interpolation, reconstruct the continuous left half outline of the face for intermediate values of y.
- By symmetry, generate the right half outline (mirror image across the vertical midline).
- Plot the complete face outline using your interpolated points.